

Ceras Csr01

Verified delivered configuration · interactive mesh available in report.html

833 km/h

DASH SPEED

7.78 h

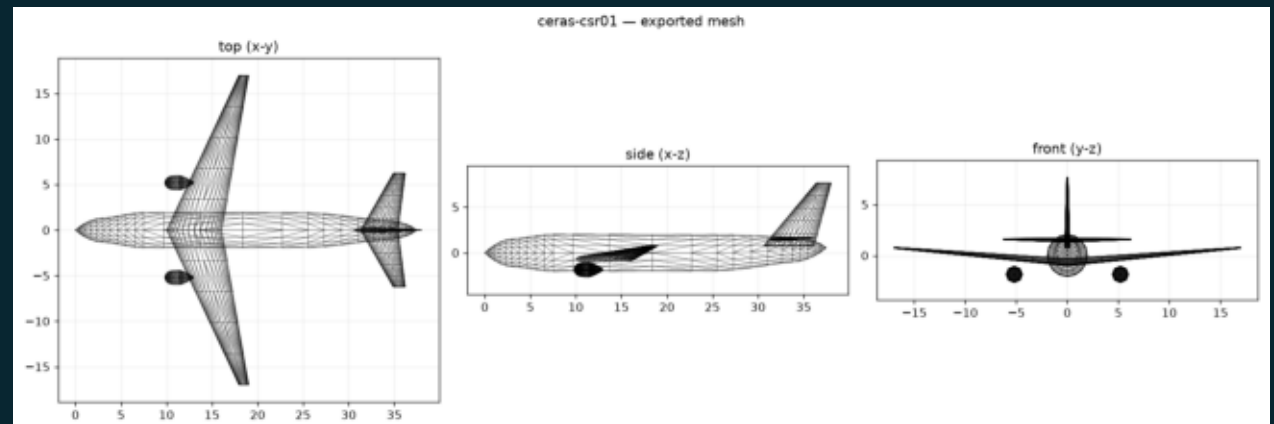
ENDURANCE

13608.0 kg

PAYLOAD

74102.0 kg

MTOW



Flight-worthiness gate review

● Schema

source=yes; optimized=yes; engine requirement=yes; payload requirement=yes; re

● Geometry truth

read-back=yes; bbox=yes; mesh checks=28

● Packing

fuel-volume margin 60.252 m³; engine/payload clear

● Balance

component-model SM full 0.283; reserve 0.276 MAC

● Pitch trim

tail incidence -2.112° vs spec -2.112°; CM residual -0.0000

● Stall

V_{stall} 73.380 m/s ≤ 75.0 m/s at CL_{max} 1.80

● Directional authority

cant-corrected fin volume V_v 0.1175 ≥ 0.02 (source-locked reproduction)

● Structures

unsupported domain: the current OAS wingbox method is qualified only for span

● Endurance & thrust

7.78 h; cruise/dash thrust ≥ buildup drag

● Reproduction closure honesty

feasible=yes; driver=yes; post-delivery verify=yes; calibration=yes; frozen=ye

● Shape fidelity

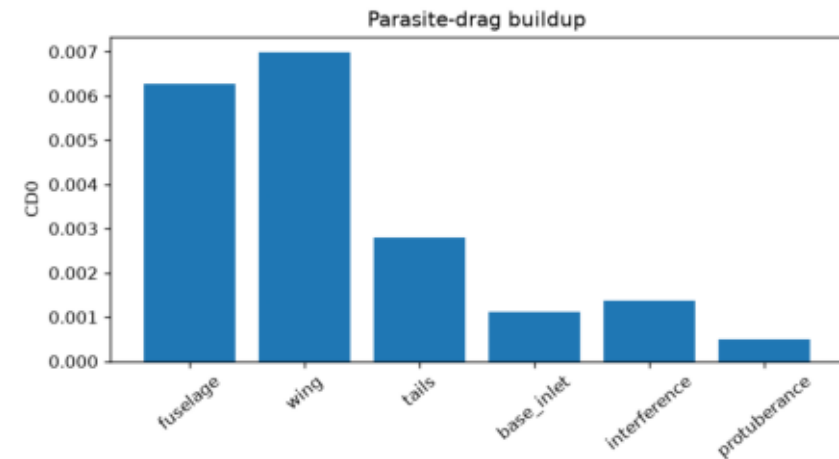
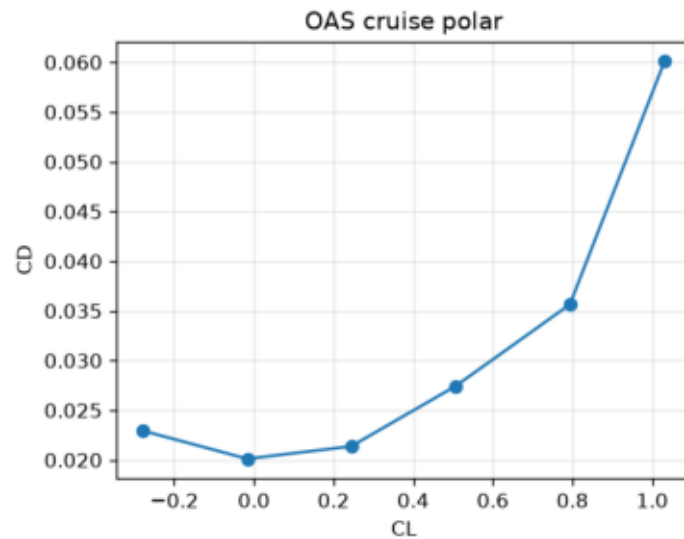
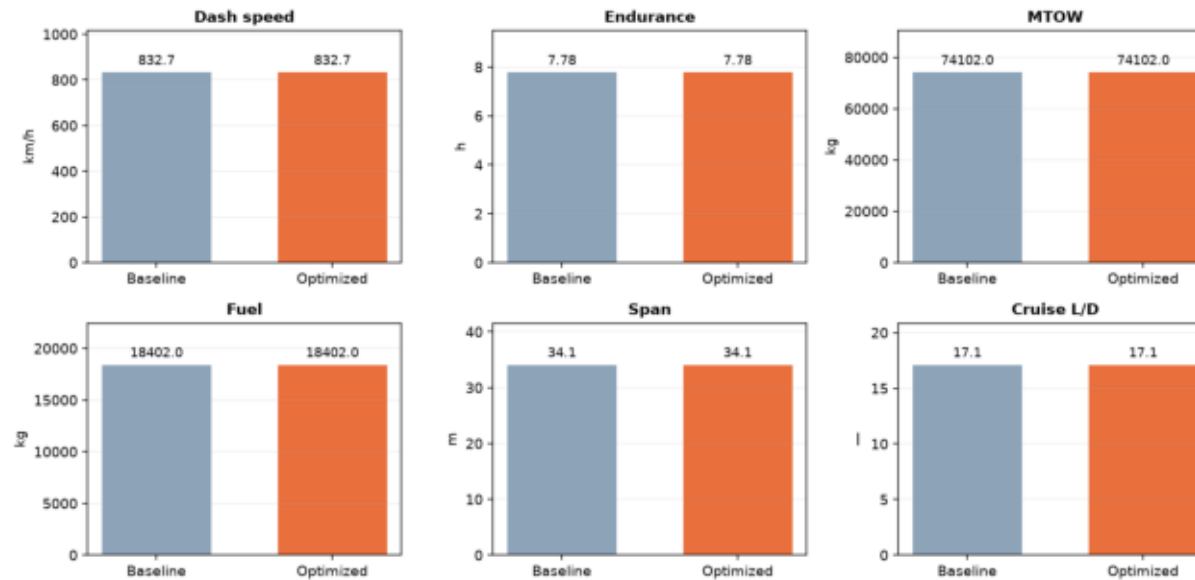
span/L 0.91; root/L 0.16; sweep 25.0°; 0 scored shape departures; 0 total audi

● Cross-checks & traceability

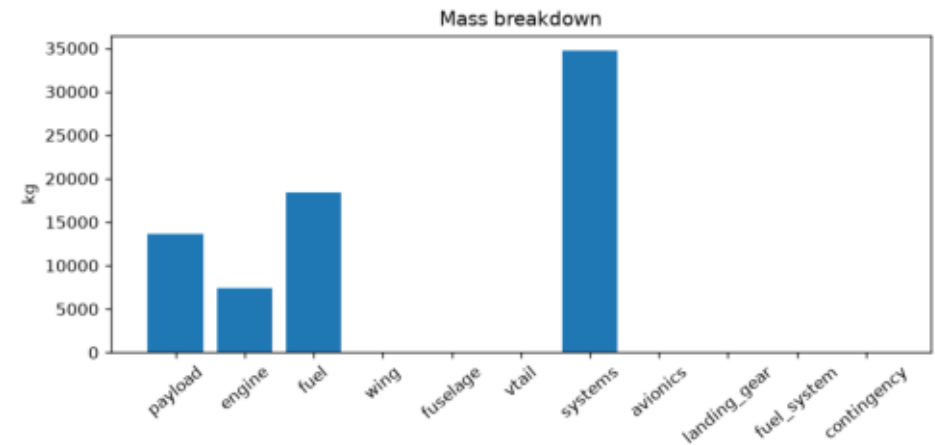
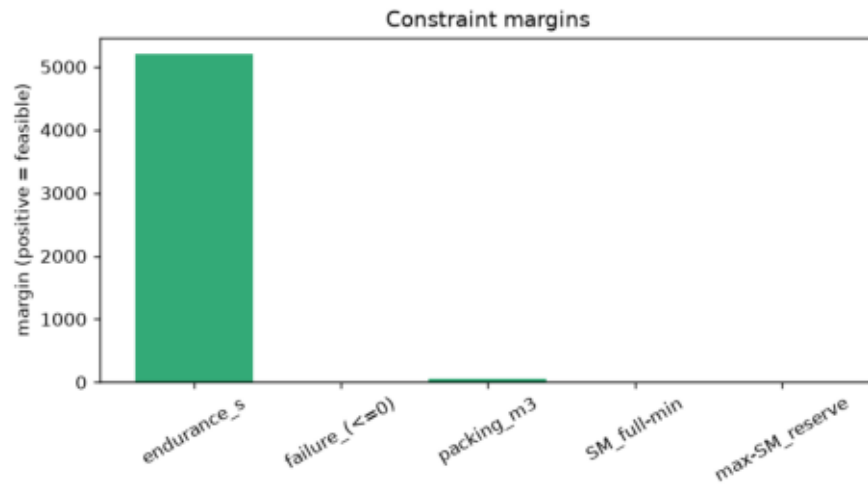
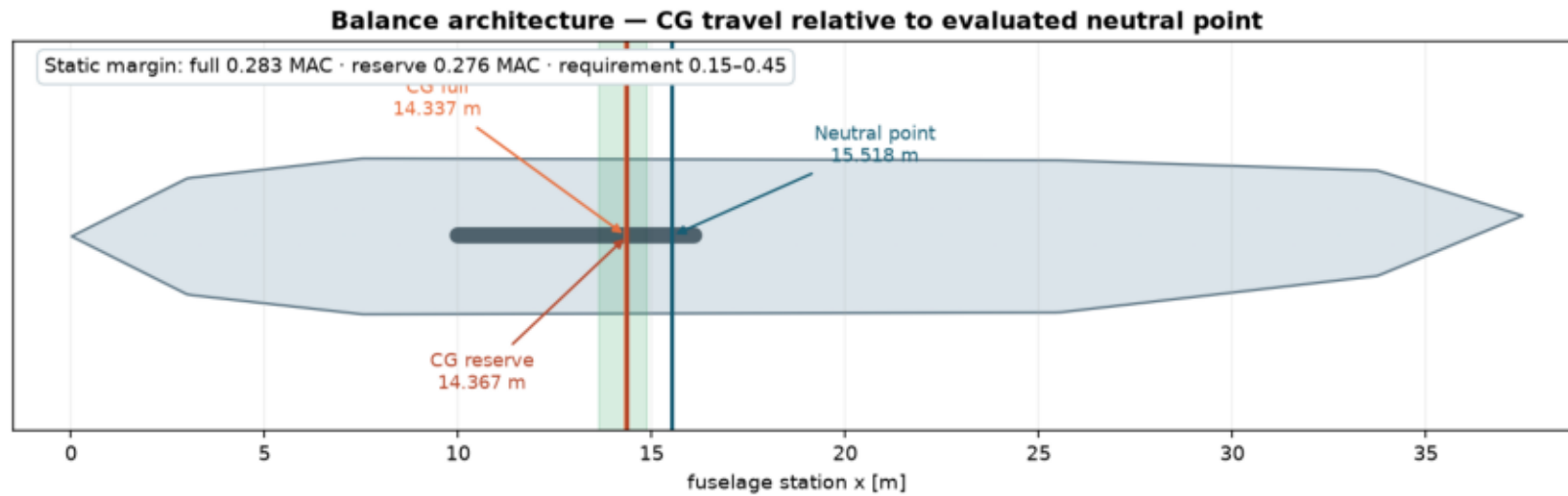
core 12/12; VSPAERO/OAS CL_α ratio 0.914; NP method spread 0.095 MAC (diagnosti

Performance and aerodynamic evidence

Design evolution: source baseline → verified optimum



Balance and structural margins



Verification summary and engineering limits

Validation core: 12/12 · all recorded checks: 14/14

- ✓ k450 tsfc kg per kgf hr
- ✓ breguet round trip
- ✓ isa T sl
- ✓ isa rho sl
- ✓ oas induced drag vs elliptic
- ✓ cantilever deflection identity
- ✓ thin airfoil cm ac
- ✓ fin volume coefficient
- ✓ fin te within body
- ✓ hybrid component provenance
- ✓ vspaero vs oas CL
- ✓ elevon cm delta vspaero vs oas
- ✓ tacs artifact schema

Documented limitations

- Engine thrust-lapse and part-throttle TSFC are assumptions anchored by one static vendor point.
- Stall uses an assumed $CL_{max}=1.80$; no high-lift or nonlinear separation model is present.
- OAS/VSPAERO are lifting-surface models; fins are gated separately by V_v and exported-mesh truth.
- SU2 is a coarse 2-D Euler calibration and must report convergence separately from successful execution.
- TACS and OAS use different wingbox idealizations; their mass/stress ratio is calibration evidence only.

Full interactive evidence, exact stage values, and configuration: report.html